

TradXSell- Test Plan Report



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Date: 12/11/2024

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Test Plan Identifier:

TP-MERN-Ecommerce-01

Introduction:

This test plan describes the testing approach for a MERN-based e-commerce website with a seller and consumer panel. While merchants manage product listings and orders, the project aims to provide a seamless platform where buyers can browse things, add items to a basket, and complete transactions. White-box, black-box, static, and automated testing are included in the testing process. To ensure thorough validation, Jest, NYC, Cypress, and Jira were used.

Scope:

This test plan's scope ensures that all functional components of the e-commerce platform are tested, thereby ensuring that the features that interact with both customers and sellers work as intended. It includes:

- Customer panel functionality testing, such as checkout, adding products to cart, and processes of browsing products.
- Seller panel features like order processing and product management are validated.
- Full-cycle testing of API interactions of the front-end and back-end systems.

Testing critical functional requirements like error handling, integration with payment gateways, and user authentication. Usability guidelines and compatibility to all major web browsers also should be ensured. These aspects are not in scope as of now. That said, load testing or performance testing or advanced mobile responsiveness aspects may be included in any subsequent test programs.

Test Items:

Application parts for testing that the following application parts include test items for these panels

Customers panel

- Product browse
- page with product detail
- Add to checkout and cart process
- Authentication and user management

Panel of Sellers:

- Products Management
- Order Management
- Management of sellers account.

General Things:

- API connectivity between the front end and the back-end.

Features to Be Tested:

Customer Panel:

- Product listing and filtering with search.
- Items putting in and taking out from cart.
- Checkout process with payment handling.
- Authentication of User using password, registration and sign in

Seller Panel:

- A product will be included after validation for example with obligatory fields.
- Order Status as processed shipped etc. .
- Checking the order details.

Must to be have Features:

- API Resolutions to CRUD operations
- Handling the errors caused due to system error or invalid inputs made.

Features Not to Be Tested

- Performance and load testing.
- A lot of concurrent user sessions are used for stress testing.

- responsiveness on mobile devices which surpasses the standards for certification.
- 3rd Parties API integration in the Project as they are not in our control.

Test Approach:

White Box Testing:

- Done to ascertain individual components and functionalities using Jest.
- Decisions and branch coverage : A coverage study was done in joint collaboration with NYC.

Black Box Testing:

- Limits of input (such as in price range or quantity). BVA is used.
- ECP is used for both Legal and invalid input cases.

Static Analysis:

- Hand carried out on a checklist to ensure that the code is compliant with standards.
- Suggested Recommendations on how to improve the system by enhancing the code quality

Automated Testing:

- End-to-end testing of 3 processes using Cypress:
 1. Order the product and Safely Checkout by customer
 2. Add the product by Seller
 3. Remove Product from the cart and Add review for any Product

Pass/Fail Criteria:

Pass:

- Every test case yields the anticipated outcomes.
- Important functionalities are working according to user requirements

Fail:

- When expected output is not equal to actual output
- When any user requirement is missing in the system

Testing Deliverables:

1. Test Plan Document.
2. Test Cases and Checklists.
3. Test Execution Logs.
4. Defect Reports (via Jira).
5. Automated Test Scripts and Results (Cypress).
6. Code Coverage Reports (NYC).

Testing Environment:

- **Development Tools:** Visual Studio Code, MongoDB Atlas.
- **Testing Tools:** Jest, NYC, Cypress, Jira.
- **Browsers:** Chrome, Firefox, Edge.
- **Server Environment:** Node.js with Express, Deployed on localhost or staging server.

Roles and Responsibilities:

- **Test Manager:** (Oversees the testing process, ensures adherence to the test plan)
Every group member did her own testing part which was assigned/divided. So everyone was the test manager for her own part.
- **Test Engineers:**
 - Perform white-box and black-box testing.
 - Execute automated tests and log results.

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- **Developers:**
 - Provide unit tests.
 - Fix identified defects.

To develop the application was not part of this project. We took fyp for our seniors. So they're developers.

- **Project Manager:** Ensures resource allocation and timelines.

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Risks and Contingencies:

Risks:

- Delays in bug fixes may impact the testing schedule.
- Limited access to real-world testing data.
- Problems specific to a particular browser were not identified during development.

Exceptions/Challenges:

- Incorporate a buffer time for defect fix in the schedule.
- When actual data is not available for testing, use dummy data.
- Provide cross-browser testing for key features as priority.

This test plan ensures a systematic approach to verifying the functionality and robustness of the MERN-based eCommerce website.